

# P3288R2

## std::elide

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## Abstract

Add a new class to the standard library to make possible the emplacement of an unmovable-and-uncopyable *prvalue* into types such as `std::optional`, `std::variant` and `boost::static_vector` -- without requiring an alteration to the definitions of these classes.

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**NOTE:** A change is required to the core language.

## § 1. Introduction

While it is currently possible to return an unmovable-and-uncopyable class by value from a function:

```
std::counting_semaphore<8> FuncReturnsByValue(unsigned const a, unsigned const b)
{
    return std::counting_semaphore<8>(a + b);
}
```

It is not possible to emplace this return value into an `std::optional`:

```
int main()
{
    std::optional< std::counting_semaphore<8> > var;

    var.emplace( FuncReturnsByValue(1,2) ); // compiler error
}
```

This paper proposes a solution to this debacle, involving an addition to the standard library, along with a change to the core language.

## § 2. Motivation

### § 2.1. `emplace( FuncReturnsByValue() )`

There is a workaround to make this possible, and it is to use a helper class with a conversion operator:

```
int main()
{
    std::optional< std::counting_semaphore<8> > var;

    struct Helper {
        operator std::counting_semaphore<8>()
        {
            return FuncReturnsByValue(1,2);
        }
    };

    var.emplace( Helper() );
}
```

This is possible because of how the `emplace` member function is written:

```
template<typename... Params>
T &emplace(Params&&... args)
{
    . . .
    ::new(buffer) T( forward<Params>(args)... );
    . . .
}
```

or:

```
template<typename... Params>
T &emplace(Params&&... args)
{
    . . .
```

```

    std::construct_at<T>( buffer, forward<Params>(args)... );
    . . .
}

```

The compiler cannot find a constructor for `T` which accepts a sole argument of type `Helper`, and so it invokes the conversion operator, meaning we effectively have the constructor of `T` invoked as follows:

```

T( FuncReturnsByValue(1,2) );

```

In this situation, where we have a *prvalue* returned from a function, we have guaranteed elision of a copy/move operation. This proposal aims to simplify this technique by adding a new class to the standard library called `std::elide` which can be used as follows:

```

int main()
{
    std::optional< std::counting_semaphore<8> > var;

    var.emplace( std::elide(FuncReturnsByValue,1,2) );
}

```

### § 3. Possible implementation

```

#include <functional>    // invoke
#include <tuple>         // apply, tuple
#include <type_traits>   // invoke_result, is_same, remove_reference, true_type
#include <utility>       // move

namespace std {
    template<typename F_ref, typename... Params_refs>
    class elide {
    public:
        typedef std::true_type tag_elide;
    private:
        using R = invoke_result_t< F_ref, Params_refs... >;
        static_assert( is_same_v< R, remove_reference_t<R> > ); // F must return Rvalue
        using F = remove_reference_t<F_ref>;
        F &&f; // 'f' is always an Rvalue reference
        tuple< Params_refs... > const args_tuple; // just a tuple full of references
    };
}

```

```

public:
    template<typename F, typename... Params>
    constexpr explicit elide(F &&arg, Params&&... args) noexcept // see
        : f(std::move(arg)), args_tuple( static_cast<Params&&>(args)... )

    constexpr operator R(void) noexcept(noexcept(apply(static_cast<F_ref>
    {
        return apply( static_cast<F_ref>(f), std::move(args_tuple) );
    }

    /* ----- Delete all miranda methods ----- */
    elide(      void      ) = delete;
    elide(elide const & ) = delete;
    elide(elide      &&) = delete;
    elide &operator=(elide const & ) = delete;
    elide &operator=(elide      &&) = delete;
    elide const volatile *operator&(void) const volatile = delete;
    template<typename U> void operator,(U&&) = delete;
    /* ----- */
};

template<typename F, typename... Params>
elide(F&&,Params&&...) -> elide<F&&,Params&&...>; // explicit deduction

template<typename T>
concept has_tag_elide_true = T::tag_elide::value;
}

```

Thoroughly tested on GodBolt: <https://godbolt.org/z/xcfo1z39K>

You can comment out **Line #85** in the godbolt to see the effect of the core language change.

## § 4. Design considerations

### § 4.1. template constructor

The above implementation of `std::elide` will not work in a situation where a class has a constructor which accepts a specialisation of the template class `std::elide` as its sole argument, such as the following `AwkwardClass`:

```
class AwkwardClass {
    std::mutex m; // cannot move, cannot copy
public:
    template<typename T>
    AwkwardClass(T &&arg)
    {
        cout << "In constructor for AwkwardClass, \n"
              "type of T = " << typeid(T).name() << endl;
    }
};

AwkwardClass ReturnAwkwardClass(int const arg)
{
    return AwkwardClass(arg);
}

int main(int const argc, char **const argv)
{
    std::optional<AwkwardClass> var;
    var.emplace( std::elide(ReturnAwkwardClass, -1) );
}
```

The above program will print out:

```
--
In constructor for AwkwardClass,
type of T = std::elide< AwkwardClass, AwkwardClass (&)(int), int&& >
--
```

The problem here is that the constructor of `AwkwardClass` has been instantiated with the template parameter type `T` set to a specialisation of `std::elide`, when really we wanted `T` to be set to `int`. We want the following output:

```
--  
In constructor for AwkwardClass,  
type of T = int  
--
```

A workaround here is to apply a constraint to the constructor of `AwkwardClass` as follows:

```
template<typename... Params>  
requires (! (1u==sizeof...(Params)) && (std::has_tag_elide_true<Params> |  
AwkwardClass(Params&&... arg)  
{  
    ( std::cout << "In constructor for AwkwardClass, type of T = " << type:  
}  
}
```

In order that class definitions do not have to be altered in order to apply this constraint to template constructors, this proposal makes a change to the core language to prevent the constructor of any class from having a specialisation of `std::elide` as its sole parameter. This constraint is achieved by detecting the presence of a `typedef` called `tag_elide` for which `tag_elide::value` evaluates to `true`.

The programmer can write their own custom class to use instead of `std::elide`, and in order to take advantage of the core language feature which prevents the instantiation of constructors, the programmer must give their custom class an accessible `typedef` as follows:

```
typedef std::true_type tag_elide;
```

Furthermore, if the programmer so wishes, they can derive their own class from `std::elide` and change the `typedef` in order to disable the failure of template instantiation:

```
template<typename F_ref, typename... Params_refs>  
class MyElide : public std::elide<F_ref,Params_refs...> {  
public:
```

```

    typedef std::false_type tag_elide;
    . . .
};

```

## § 5. Alternatives

### § 5.1. `emplace_invoke`

An alternative would be to add a method called `emplace_invoke` to classes like `std::optional` and `std::variant`, as follows:

```

template<typename F, typename... Params>
T &emplace_invoke(F &&f, Params&&... args)
{
    . . .
    ::new(buffer) T( std::forward<F>(f)( std::forward<Params>(args)... ) );
    . . .
}

```

The benefit of `emplace_invoke` is that it doesn't require a change to the core language. The benefit of `std::elide` is that pre-existing header files which implement or use the `emplace` method can be used with `std::elide` without requiring an alteration to the header file. (For example, there would be no need to alter the *Boost* header file for `boost::static_vector` in order to use it with `std::elide`).

## § 6. Proposed wording

The proposed wording is relative to [\[N4950\]](#).

In subclause 13.10.3.1.11 [\[temp.deduct.general\]](#), append a paragraph under the heading *"Type deduction can fail for the following reasons:"*

- 11 -- Attempting to instantiate a constructor that has exactly one parameter and the sole parameter is a class type which contains an accessible typedef called 'tag\_elide' whose 'value' evaluates to true, for example

```
typedef std::true_type tag_elide;
```

## § 7. Impact on the standard

This proposal is a library extension combined with a change to the core language. The change to the core language is a paragraph to be added to 13.10.3.1.11 [[temp.deduct.general](#)]. The addition has no effect on any other part of the standard.

## § 8. Impact on existing code

No existing code becomes ill-formed. The behaviour of all existing code is unaffected.

## § 9. Revision history

R1 => R2

- New section entitled '*Alternatives*' to suggest that `std::optional` and `std::variant` could be given an `emplace_invoke` method.
- `std::elide` can be used in a constant-evaluated context.
- `std::construct_at` is given as an alternative to *placement new* (note that the former can be used in a constant-evaluated context).
- The example of `std::any` is removed because `T` must be copy-constructible.
- The example of `boost::static_vector` is added.
- The proposed wording for the standard clarifies that the accessible typedef is called `tag_elide`
- Correction of typo, `is_elider` -> `has_tag_elide_true`
- Correction of typo, `typename T` -> `typename... Params`

R0 => R1

- The class `std::elide` is not defined as `final`
- Template instantiation only fails if the constructor has exactly one parameter.
- Template instantiation fails based upon the presence of a `typedef` tag whose `value` evaluates to `true`.

## § 10. Acknowledgements

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## § References

### § Normative References

[N4950]

Thomas Köppe. *Working Draft, Standard for Programming Language C++*. 10 May 2023. URL: <https://wg21.link/n4950>